SQL Data Analysis Using Superstore Dataset

Introduction

SQL (Structured Query Language) is one of the most widely used tools for managing and analyzing data. It helps in retrieving, filtering, grouping, and summarizing data efficiently. The objective of this project is to perform data analysis on the Superstore dataset using SQL queries and extract meaningful business insights.

Objectives

- Understand the structure of the dataset.
- Perform data analysis using SQL queries.
- Apply SQL functions such as COUNT, SUM, and AVG.
- Use filtering and sorting techniques.
- Generate meaningful insights from sales data.

Tools and Technologies Used

- Python
- Pandas
- SQLite (pandasql)
- Google Colab

Dataset Description

The Superstore dataset contains information related to customer orders, sales transactions, shipping details, customer segments, and product categories. The dataset is useful for analyzing sales performance and customer behavior.

SQL Queries Performed

1. Display Sample Records

Purpose:
To view the first few records of the dataset and understand its structure.

2. Count Total Records

SQL Query Used:
COUNT()

```
pysqldf(query)
```

	Segment	Total_Customers
0	Consumer	5101
1	Corporate	2953
2	Home Office	1746

Purpose:
To determine the total number of records present in the dataset.

3. Customer Analysis by Segment

SQL Query Used:
GROUP BY, COUNT()

```
pysqldf(query)
```

	Segment	Total_Customers
0	Consumer	5101
1	Corporate	2953
2	Home Office	1746

Purpose:
To identify the number of customers/orders in each segment.

4. Average Sales by Segment

SQL Query Used:
AVG(), GROUP BY

	Segment	Average_Sales
0	Consumer	225.065777
1	Corporate	233.150720
2	Home Office	243.403309

Purpose:
To calculate the average sales generated by each customer segment.

5. Total Sales by Category

SQL Query Used:
SUM(), GROUP BY, ORDER BY

	Category	Total_Sales
0	Technology	827455.8730
1	Furniture	728658.5757
2	Office Supplies	705422.3340

Purpose:
To identify which product categories contribute the highest sales.

6. Top Sales Records

SQL Query Used:
ORDER BY DESC, LIMIT

	Customer Name	Sales
0	Sean Miller	22638.480
1	Tamara Chand	17499.950
2	Raymond Buch	13999.960
3	Tom Ashbrook	11199.968
4	Hunter Lopez	10499.970
5	Adrian Barton	9892.740
6	Sanjit Chand	9449.950
7	Bill Shonely	9099.930
8	Sanjit Engle	8749.950
9	Christopher Conant	8399.976

Purpose:
To identify the highest-value sales transactions.

7. High Value Orders

SQL Query Used:
WHERE

... w ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales
1	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032.0	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	1706.184
5	CA-2016-106320	25/09/2016	30/09/2016	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057.0	West	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	1044.630
8	US-2016-150630	17/09/2016	21/09/2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140.0	East	FUR-BO-10004834	Furniture	Bookcases	Riverside Palais Royal Lawyers Bookcase, Royal...	3083.430
6	CA-2017-117590	08/12/2017	10/12/2017	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080.0	Central	TEC-PH-10004977	Technology	Phones	GE 30524EE4	1097.544
5	CA-2017-105816	11/12/2017	17/12/2017	Standard Class	JM-15265	Janet Molinari	Corporate	United States	New York City	New York	10024.0	East	TEC-PH-10002447	Technology	Phones	AT&T CL83451 4-Handset Telephone	1029.950

8	CA-2015-106376	05/12/2015	10/12/2015	Standard Class	BS-11590	Brendan Sweed	Corporate	United States	Gilbert	Arizona	85234.0	West	OFF-AR-10002671	Office Supplies	Art	BOSTON Model 1606 High-Volume Electric Pe...	1113.024
0	CA-2017-114489	05/12/2017	09/12/2017	Standard Class	JE-16165	Justin Ellison	Corporate	United States	Franklin	Wisconsin	53132.0	Central	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	1951.840
6	CA-2015-139892	08/09/2015	12/09/2015	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207.0	Central	TEC-MA-10000822	Technology	Machines	Lexmark MX611dhe Monochrome Laser Printer	8159.952
8	CA-2015-139892	08/09/2015	12/09/2015	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207.0	Central	FUR-CH-10004287	Furniture	Chairs	SAFCO Arco Folding Chair	1740.060
6	CA-2016-146262	02/01/2016	09/01/2016	Standard Class	VW-21775	Victoria Wilson	Corporate	United States	Medina	Ohio	44256.0	East	TEC-MA-10000864	Technology	Machines	Cisco 9971 IP Video Phone Charcoal	1188.000

Purpose:
To filter orders having sales greater than a specified value.

Results and Observations

1. The dataset contains sales records from multiple customer segments.
2. Customer segments contribute differently to overall sales.
3. Product categories show variation in sales performance.
4. High-value sales transactions can be identified using filtering conditions.
5. Aggregation functions such as COUNT, SUM, and AVG help summarize business performance.
6. SQL makes it easier to extract useful insights from large datasets.

Key Findings

- Customer segments have different purchasing patterns.
- Certain product categories generate higher sales than others.
- High-value orders contribute significantly to total revenue.
- SQL queries help in organizing and analyzing business data efficiently.

Conclusion

This project demonstrated the use of SQL for data analysis. By applying various SQL operations such as SELECT, WHERE, GROUP BY, ORDER BY, COUNT, SUM, and AVG, meaningful insights were extracted from the Superstore dataset. The project enhanced practical knowledge of SQL and its role in data analytics and business decision-making.

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